Low-Precision Execution and Speculative Decoding

Deep Dive into Kimi K3 — Chapter 46

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Foundation Books Project

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Goal: bind the deployed numerical contract, verify drafts exactly against it, commit and recover every state object at one boundary, and count all the work before claiming a speedup — each claim in its evidence class.

Roadmap

Precision defines the target

Exact speculation

State and cost

Closing the book

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The deployed numerical contract

- MX blocks: 32 elements plus an E8M0 scale; MXFP4 is 4.25 bits per value. Fixture M44-A reconstructs $(1, 3, 6, 12)$ and a represented dot product of 1 versus real 1.4. [DERIVED: book fixture]

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- Algorithmic exactness preserves the deployed law p^{χ}; parity with a reference is separate — TV .02 can still flip the greedy token.

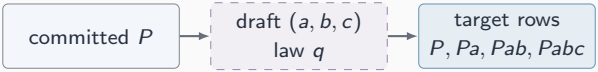
Exact speculation

Draft, verify exactly, commit a prefix



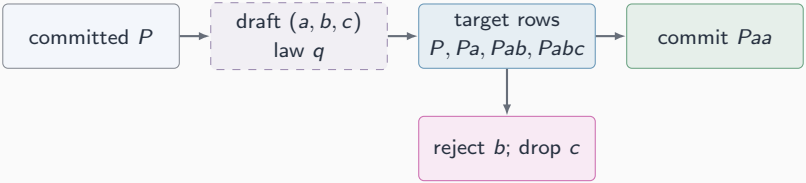
- Accept with $\min(1, p/q)$; on rejection draw from $(p - q)_+/Z$. R44-A: $\alpha = .80$, correction law $(.75, 0, .25, 0)$. [DERIVED: book fixture R44-A]

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- Accepted tokens form a prefix of one branch; the target supplies a correction or, after full acceptance, a bonus.

State and cost

One boundary for every state object; one ledger for all work

- KV keeps committed rows; KDA restores and replays (scalar $2 \to 5 \to 11$, never 27) or reconstructs directly from cached corrections. Draft, parser, random stream, and batch row move as one bundle.

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Diagnose in dependency order: identity, causality, exactness, commit, recovery, alignment, then performance.

Closing the book

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One discipline runs through all three volumes: name the contract, keep every claim in its evidence class, and check the smallest discriminating case first. Thank you for reading.